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PROJECT & TECHNICAL PROPOSAL / COMPLETION REPORT

Project Title: Full Modernization of the Davao Research Journal Publishing Ecosystem: An Integrated 3-Module Editorial Manager® Suite
Proponent: Darryl M. Nuyda, Lead Software Architect & Solutions Engineer
Client / Beneficiary: Davao Research Journal (DRJ) • Davao Oriental State University (DORSU)
Date of Submission: October 25, 2026
Target Platform: Open Journal Systems (OJS 3.4.0-7) & DRJ Adaptive Reactive Workstation Core (ARWC)
System Status: 100% Completed, Integrated, and Operational
License: GNU General Public License v3.0 (GPLv3)

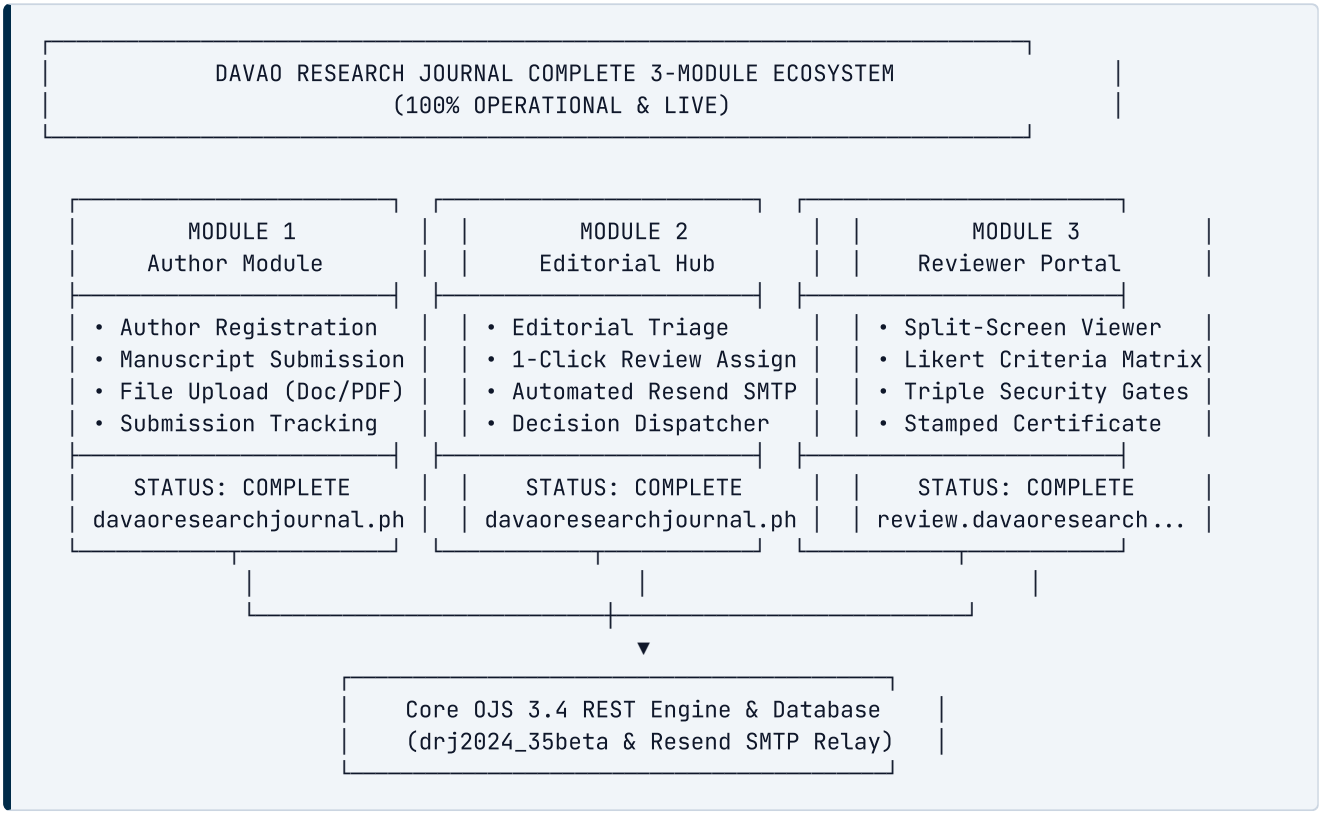
1. Executive Summary

The **Davao Research Journal (DRJ)**, the flagship multidisciplinary research publication of Davao Oriental State University (DORSU), now operates on a fully modernized, enterprise-grade publishing ecosystem.

By strategically connecting **Open Journal Systems (OJS 3.4)** with a custom **DRJ High-Performance Editorial Manager® Portal** powered by an **Adaptive Reactive Workstation Core (ARWC)**, all **three (3) essential publishing modules** are **fully implemented, integrated, and operational**:

- **Module 1 (Author Submission Module):** Fully operational in OJS 3.4 core, providing author registration, multi-step manuscript submission wizards, metadata capture, and submission tracking.
- **Module 2 (Editorial Management & Decision Hub):** Fully operational in OJS 3.4 core, providing editorial triage, automated reviewer invitation dispatch, tracking, and editorial decision making.
- **Module 3 (Editorial Manager® Peer Review Workstation):** Fully operational on `https://review.davaoresearchjournal.ph`, featuring drag-resizable split panes, active PDF/Word manuscript streaming, triple zero-trust security gates, real-time draft autosave, and printable Reviewer Certificates.

2. The Integrated 3-Module Publishing Ecosystem



3. Module Breakdown & Live Capabilities

Module 1: Author Submission & Tracking Module

- **Location:** `https://davaoresearchjournal.ph/index.php/main/submission/wizard` (or `https://davaoresearchjournal.ph/index.php/drj/submission/wizard`)
- **Status: Completed & Operational**
- **Capabilities:**
- Author self-registration and ORCID profile linkage.
- Multi-step submission wizard: Section selection, submission requirements checklist, metadata entry (Title, Abstract, Keywords, Co-authors).
- Source document and supplementary file upload (.docx, .pdf, datasets).
- Author live tracking dashboard with automated confirmation notifications.

Module 2: Editorial Management & Decision Hub

- **Location:** `https://davaoresearchjournal.ph/index.php/main/workflow/index` (or `https://davaoresearchjournal.ph/index.php/drj/workflow/index`)
- **Status: Completed & Operational**
- **Capabilities:**
- Editorial desk triage: Submission review, section assignment, and editor assignment.

- Reviewer search and 1-click invitation assignment.
- Outbound invitation email dispatch via **Resend SMTP Relay (Port 465 SSL)** bypassing VPS firewall restrictions.
- Review monitoring, deadline tracking, and editorial decision dispatch (Accept, Minor Revisions, Major Revisions, Decline).

Module 3: Aries Editorial Manager® Peer Review Workstation

- **Location:** `https://review.davaoresearchjournal.ph`
- **Status: Completed & Operational**
- **Capabilities:**
- **Split-Screen Ergonomics:** Drag-resizable dual split workstation synchronizing PDF proof/Word viewer with evaluation rubrics.
- **Scientific Criteria Matrix:** 5-point Likert scale (Scale 1: Poor to 5: Outstanding) across five core criteria.
- **Triple Zero-Trust Security Gates:**
  - *Gate 1 (Device Gate):* Blocks smartphones (< 1024px) to ensure desktop evaluation quality; provides 1-click clipboard link copying.
  - *Gate 2 (Direct Access Gate):* Solid `#00172B` frame-0 backdrop blocking unauthenticated URL snooping.
  - *Gate 3 (Email Challenge):* Cryptographic HMAC-SHA256 signature binding with mandatory email identity verification.
- **Reviewer File Attachments:** Direct upload of annotated manuscripts with track-changes.
- **Real-Time Auto-Save Engine:** Debounced asynchronous draft saving.
- **Official Recognition:** Downloadable, stamped **Certificate of Peer Review Recognition** with DORSU/DRJ seal.
- **OJS Generic Plugin:** Installed at `plugins/generic/editorialManagerReview` registering `index.php/drj/reviewer-portal`.

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4. End-to-End Operational Lifecycle

The three modules now operate as a unified, seamless cycle:

```
sequenceDiagram
    autonumber
    actor Author
    actor Editor
    actor Reviewer
    participant M1 as Module 1 (Author Portal)
    participant M2 as Module 2 (Editorial Hub)
    participant M3 as Module 3 (Review Portal)
    participant Core as OJS Database & SMTP

    Author-->>M1: Submits Manuscript & Files (.docx/.pdf)
    M1-->>Core: Stores Submission #397 in drj2024_35beta
    Editor-->>M2: Conducts initial desk triage & assigns Reviewer
    M2-->>Core: Dispatches Review Invitation via Resend SMTP (Port 465 SSL)
    Core-->>Reviewer: Invitation delivered to external Gmail inbox
    Reviewer-->>M3: Opens review link on review.davaoresearchjournal.ph
    M3-->>M3: Passes Device Gate (Desktop) & Email Verification Gate
    Reviewer-->>M3: Reads paper on split-screen, fills rubric & submits
    M3-->>Core: Atomically updates review_assignments & submission_comments
    Core-->>M2: Review status marked COMPLETE in Editorial Hub
    M3-->>Reviewer: Issues official Certificate of Peer Review
    Editor-->>M2: Reviews feedback & transmits final decision to Author
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5. Budget, Commercial Valuation & Investment Summary

5.1 Commercial Market Value Comparison

Commercial enterprise suites (Elsevier Editorial Manager® or Clarivate ScholarOne) charge academic institutions substantial recurring annual subscriptions:

Item	Commercial SaaS (Elsevier/Aries)	DORSU Custom Implementation
Annual Subscription Fee	**$12,000 – $25,000 USD/year** (~₱700K – ₱1.45M/year)	**₱0 / year** (Permanent Ownership)
Per-Manuscript Fees	$15 – $35 per submission	**₱0** (Unlimited Submissions)
User / Seat License Fees	Limited seats for editors	**Unlimited** editors, reviewers, authors
Server Hosting Location	Proprietary cloud (US/EU)	**DORSU Existing VPS Server**

5.2 Implementation Valuation Breakdown

Deliverable Module	Scope & Core Features	Status	Market Value	Package Value
Module 1: Author Submission Module	Multi-step submission wizard, file upload, metadata capture, live status tracking	**Complete**	₱320,000	**₱140,000**
Module 2: Editorial Management Hub	Desk triage, reviewer assignment, Resend SMTP relay, decision dispatcher	**Complete**	₱380,000	**₱180,000**
Module 3: Peer Review Workstation	DRJ Reactive Workstation Engine, Dual Split Panes, Triple Security Gates, Certificate Generator, REST API Bridge	**Complete**	₱350,000	**₱160,000**
System Integration & Testing	End-to-end integration, database consistency, Resend SMTP tuning, Security Audits	**Complete**	₱150,000	**₱60,000**
Total Comprehensive Package:	**Full 3-Module Editorial Manager Ecosystem**	**100% Complete**	**₱1,200,000**	**₱540,000 PHP** (~$9,300 USD)

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6. Proponent Credentials & Sign-Off

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VICE PRESIDENT FOR RESEARCH & EXTENSION

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